Lanchester Models of the Ardennes Campaign

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A detailed data base of the Ardennes campaign of World War II (December 15, 1944 through January 16, 1945) has recently been developed. The present article formulates four Lanchester models of the campaign and estimates their parameters for these data. Two-sided time histories of warfare on battles and campaigns are very rare, so Lanchester models have seldom been validated with historical data. The models are homogeneous in that tanks, armored personnel carriers, artillery, and manpower are weighted to yield a measure of strength of the Allied and German forces. This weighting is utilized for combat power and for losses. The models treat combat forces in the campaign (including infantry, armor, and artillery manpower) and total forces in the campaign (including both combat manpower and support manpower.) Four models are presented. Two models have five parameters (Allied individual effectiveness, German individual effectiveness, exponent of shooting force, exponent of target force, and a tactical parameter reflecting which side is defending and attacking.) The other two models remove the tactical parameter, which is not generally known prior to warfare, and estimate the other parameters without the tactical parameter. The main results of the research are (a) the Lanchester linear model fits the Ardennes campaign data in all four cases, and (b) when combat forces are considered Allied individual effectiveness is greater than German individual effectiveness, whereas when total forces are considered Allied and German individual effectiveness is the same. The interpretation of the latter result is that the two sides had essentially the same individual capabilities but were organized differently—the Allies chose to have more manpower in the support forces, which yielded greater individual capabilities in the combat forces. The overall superiority of the Allies in the campaign led to the attrition to the Allies being a smaller portion of their forces.
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INTRODUCTION

The objective of this research is to fit Lanchester equations to Ardennes campaign data. The two main areas of interest are the quality of the fits and the insights provided by the equations. The source of the data is the Ardennes Campaign Simulation Data Base (ACSDB), documented in Data Memory Systems, Inc. [9].

The data are newly available. They were developed by Data Memory Systems, Inc. under contract with the U.S. Army Concepts Analysis Agency. The data are two-sided, time phased (daily), and detailed. The present analysis treats tanks, armored personnel carriers (APCs), artillery, and manpower.

The data cover 33 days of the campaign, from December 15, 1944 through January 16, 1945. However, the data for the first day seem to be incomplete on the German side. The Germans attacked during days 1–6 and the Allies attacked during days 7–33. Preliminary research has shown that a form of the model that includes specification of which side is attacking and defending fits the data significantly better than if this factor is left out of the model. Also, the heaviest attrition takes place at the beginning of the campaign. On the

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basis of these considerations, the analysis treats data for days 2–11, or five days during which each side is attacking—2–6 for the Germans and 7–11 for the Allies.

In addition to the data base utilized in the present investigation there is an earlier-developed set of Ardennes campaign data, presented and analyzed in Cockrell and Carter [5], and further analyzed in Goad [12]. In this supplementary data base there are 15 two-sided observations on battle days that include complete data on forces and attrition. Six of these battle days occur between days 2 and 11. Analysis of these data has tended to confirm the present results, but this should not be thought of as a validation.

The data from Data Memory Systems, Inc. [9] give a daily record of the Ardennes campaign. The data are unique in that a time history of forces on both sides, and casualties on both sides, is included in the data base. Lanchester equations have been fit to warfare data by Engel [11] and the results confirmed with minor revisions by Samz [25]. However, time histories of other combat situations have been very difficult to obtain, and thus Lanchester models have been difficult to validate. Helmbold [20] presents a convincing argument for the need for time histories of warfare. He discusses the work of Willard [33], which used initial and final forces data from Bodart [1]. He concludes, on the basis of an analysis of the data of Engel [11], that various forms of Lanchester equations could fit the data as well as those chosen. The present article formulates a general equation form that can result in Lanchester square, Lancester linear, or something else. The results of this analysis of the Ardennes campaign tend to be close to Lanchester linear; other campaigns could, of course, have different characteristics and yield different results.

Lanchester equations fit to within-war two-sided data have been very rare due to the nonavailability of such data. Thus the present investigation is of interest in that regard. In the future, two-sided data may be more available. Methodology for the analysis of such data will become an important topic; this analysis should be thought of as a prototype.

HISTORICAL OVERVIEW

The following historical overview is extracted from Data Memory Systems, Inc. [9]:

On December 16, 1944 three German Armies—the Fifth Panzer (Armored), Sixth Panzer, and Seventh—launched a surprise offensive against a thinly held section of the U.S. front line in rugged, wooded terrain centered around the intersection of the borders of Luxembourg, Germany, and Belgium. This area, known as the Ardennes, was to become the focal point of a bitterly contested battle of several weeks' duration fought between units of the three German armies mentioned above and units of the U.S. First and Third Armies and the British XXX Corps.

The Ardennes Campaign, popularly known as the Battle of the Bulge, caught the U.S. units by almost complete surprise. By late 1944, the Allied High Command were confident that the Germans lacked the materiel and personnel resources to launch a major attack against Allied forces on the Western Front. However, under cover of strict security measures and a clever deception plan, the Germans managed to assemble an impressive force of armored, artillery, and infantry units without detection by the Allies. The overly ambitious German plan called for a penetration of the Allied lines in the same area used by the Germans in their victorious campaign in 1940, a swift advance to the port city of Antwerp which would split the British and U.S. forces and defeat the Allied armies in detail. Although impressive on paper, the forces assembled for the German attack suffered from a number of major weaknesses. Personnel quality was low, transportation networks were inadequate, and German air forces, though massed for the offensive, were wholly inade-

quate in the face of the overwhelming air supremacy of the Allies. Nevertheless, aided by inclement weather conditions, which grounded U.S. aircraft, the Germans massed forces and penetrated the U.S. line in several places, inflicting severe defeats on two U.S. divisions in the opening phases of the battle.

After several days of German penetrations, U.S. forces rallied to slow and then to stop the German attack. In fact, on the evening of the first day of the battle, U.S. units on the northern shoulder of what was to become the bulge—the salient created by the German attack—had blunted critical attacks by some of the most formidable of the German panzer units, which had been ordered to reach the Meuse River line on the way to Antwerp by the end of the first day of the attack. Hard-fought defensive operations by U.S. units at St. Vith and by the U.S. 101st Airborne Division also hindered the German advance. From the south, by beginning about December 18, Lieutenant General George S. Patton's U.S. Third Army attacked into the left shoulder of the bulge to relieve beleaguered U.S. units at Bastogne. By Christmas Day, skies had cleared over the Ardennes region and the full might of Allied air supremacy was brought to bear on the German units in the bulge; the Allies counterattacked, and approximately two weeks later restored the front line in the Ardennes to its pre-December 16 positions.

MODELS

1. General Form of the Model

The general form of the model to be estimated is

$$\dot{B} = a\,(d \text{ or } 1/d)R^p B^q$$

$$\dot{R} = b\,(1/d \text{ or } d)B^p R^q,$$

where

B, R = Blue forces, Red forces,
$\dot{B}, \dot{R}$ = Blue forces killed, Red forces killed,
a, b = Red kills Blue attrition parameter, Blue kills Red attrition parameter,
d or $1/d$ = tactical parameter—factor for attrition to defender (d) or attrition to attacker ($1/d$),
p = exponent parameter of shooting force,
q = exponent parameter of target force,

Data are available daily, so $\dot{B}$ and $\dot{R}$ are kills per day.

Blue denotes the Allies (United States and Great Britain) and Red denotes the Germans. Forces are weighted tanks, APCs, artillery, and manpower.

The parameters p and q determine how close the data conform to the Lanchester square or Lanchester linear models. If $p = 1$ and $q = 0$ the Lanchester square model holds. If $p = 1$ and $q = 1$ the Lanchester linear model holds.

Lanchester [23] proposed the Lanchester square model of modern warfare, in which all shooters are capable of concentrating their fire on all targets. This he contrasted with ancient warfare, in which essentially all combat is comprised of one-on-one encounters, and which is captured by the Lanchester linear model.

In the Ardennes campaign, to the extent that both sides concentrate their forces, the Lanchester square equations should be expected to hold. If, however, the campaign is best

characterized as a collection of small engagements where the quantitatively superior side is not usually locally superior, the Lanchester linear equations should hold.

The parameter d is a multiplier of attrition due to being a defender. If $d < 1$ then the defender receives less casualties due to being a defender (there is a defender advantage). If $d > 1$ then the defender receives more casualties due to being a defender (there is an attacker advantage).

2. Model 1—Combat Forces

In Model 1 forces are composed of tanks, APCs, artillery, and combat manpower, where combat manpower is made up of infantry, armor, and artillery personnel. Manpower casualties are defined as killed and wounded. Forces are tanks, APCs, artillery, and combat manpower weighted by 20, 5, 40, and 1, respectively.

3. Model 2—Total Forces

In Model 2 forces include all personnel in the campaign. Logistics and support personnel of all types are counted. Casualties are defined as killed, wounded, captured/missing in action, and disease and nonbattle injuries. In the Ardennes campaign, as in other campaigns in World War II, the Allies had a smaller portion of their forces in combat, as opposed to logistics and support, than their German opponents.

In estimating the parameters of Model 1 it is found that Red effectiveness a is less than Blue effectiveness b; these parameters are for combat forces only. This should not be surprising, however, because the Red combat forces have less support and thus should not be as individually effective. In Model 2, when all personnel are included, it is found that individual effectiveness is similar for both sides, for the technology and societal characteristics of the opposing forces were similar; the sides simply chose to organize their forces differently.

4. Model 3—Combat Forces, No Tactical Parameter

Definitions are the same as in Model 1, but the parameter d, which requires knowledge of which side is defending, is not estimated. From the point of view of force structure analysis before the fact, where one may not be able to rely on being either the defender or the attacker, it is important to know if the estimate of the exponents p and q depends upon knowing the tactical parameter d.

5. Model 4—Total Forces, No Tactical Parameter

Model 4 is similar to Model 3 in that d does not appear. Model 4 addresses total forces rather than combat forces.

6. Discussion of Lanchester Models

Lanchester [23] is the basic document that proposed the Lanchester models. Taylor [29] is an extensive discussion of the mathematical characteristics of Lanchester processes (see also Taylor [28].). U.S. Army Material and Readiness Command [31] contains a thorough treatment of the theory of Lanchester models.

Table 1. Data on tanks.

Day	Blue tanks	Blue tanks killed	Red tanks	Red tanks killed
1	2853.	1.	0.	0.
2	2863.	12.	747.	10.
3	2867.	43.	663.	7.
4	2840.	60.	639.	13.
5	2808.	64.	669.	21.
6	3965.	33.	619.	11.
7	4082.	10.	595.	21.
8	4109.	15.	615.	5.
9	4086.	36.	645.	24.
10	4062.	48.	596.	22.
11	4265.	24.	544.	28.
12	4520.	20.	483.	14.
13	4511.	19.	466.	13.
14	4526.	18.	450.	7.
15	4541.	16.	433.	7.
16	4516.	20.	428.	21.
17	4610.	10.	403.	5.
18	4695.	14.	413.	9.
19	4701.	24.	419.	6.
20	4710.	26.	431.	2.
21	4728.	22.	428.	12.
22	4686.	13.	394.	2.
23	4719.	13.	396.	2.
24	4684.	12.	400.	8.
25	4703.	9.	407.	0.
26	4743.	7.	398.	7.
27	4761.	5.	407.	2.
28	4745.	7.	407.	2.
29	4717.	2.	393.	0.
30	4699.	6.	418.	13.
31	4678.	16.	410.	5.
32	4662.	11.	434.	3.
33	4628.	20.	432.	2.

Dolansky [10] presented a survey of the state of the art of Lanchester theory up to that time. Helmbold [15] discussed predictions of warfare outcomes in terms of the theory, addressing in particular the predictions of Engel [11] and referring to analyses of data contained in Helmbold [14]. Further analyses of data appeared in Helmbold [16]. Helmbold [17] proposed a general attrition equation, with three parameters.

Models 1 and 2 above have five parameters to be estimated, and Models 3 and 4 above have four parameters to be estimated. The parameters a and b are in Helmbold's general model. The parameters p and q are estimated separately here, because from an a priori point of view it is desired to allow p and q to both vary freely between 0 and 1. The tactical parameter d is a bonus of the present analytical effort. It significantly improves the fit and has a subjectively plausible explanation. Estimates are also made without it, however, because it is not known in advance by force structure planners.

DATA

Tables 1–4 present data on tanks, APCs, artillery, and combat manpower for days 1–33 of the Ardennes campaign (December 15, 1944 through January 16, 1945).

Table 2. Data on APCs.

Day	Blue APCs	Blue APCs killed	Red APCs	Red APCs killed
1	6103.	0.	0.	0.
2	6019.	33.	2046.	5.
3	5970.	46.	2041.	20.
4	5908.	18.	2021.	12.
5	6004.	37.	2009.	18.
6	7274.	11.	1984.	22.
7	7295.	6.	1952.	19.
8	7507.	13.	2065.	16.
9	7533.	6.	2034.	34.
10	7486.	18.	1970.	20.
11	8105.	2.	1875.	36.
12	8552.	2.	1800.	31.
13	8629.	3.	1731.	31.
14	8536.	3.	1659.	22.
15	8552.	2.	1595.	15.
16	8565.	0.	1542.	7.
17	8554.	2.	1532.	14.
18	8615.	0.	1523.	8.
19	8593.	0.	1516.	10.
20	8462.	0.	1451.	10.
21	8578.	0.	1441.	12.
22	8564.	2.	1419.	6.
23	8502.	0.	1409.	5.
24	8375.	0.	1403.	16.
25	8418.	0.	1364.	2.
26	8446.	0.	1360.	3.
27	8476.	0.	1358.	9.
28	8348.	1.	1349.	8.
29	8459.	1.	1341.	3.
30	8454.	0.	1335.	17.
31	8374.	0.	1322.	6.
32	8436.	0.	1318.	4.
33	8363.	0.	1309.	3.

Table 5 presents data on combat forces, where the data from Tables 1–4 on tanks, APCs, artillery, and combat manpower are weighted by 20, 5, 40, and 1, respectively.

Table 6 presents data on total manpower. Table 7 presents data on total forces.

The basic distinctions between combat manpower and total manpower are that combat manpower includes only infantry, armor, and artillery personnel, and total manpower includes these categories plus maintenance (two types), medical, transportation, supply, engineering, and other. Combat manpower casualties are defined as killed and wounded, and total manpower casualties are defined as killed and wounded plus captured/missing in action and disease and nonbattle injuries.

Aggregation to the combat forces of Table 5 and the total forces of Table 7 reflect a particular aggregation which is a limitation of the analysis. The weights given above are consistent with those of studies and models of the U.S. Army Concepts Analysis Agency. Virtually all theater-level dynamic combat simulation models incorporate similar weights, either as inputs or as decision parameters computed as the simulations progress. Such weights are also utilized in aggregate policy studies; for instance, of particular interest among studies using linear weights are Congressional Budget Office [7, 8], addressing

Table 3. Data on artillery.

Day	Blue artillery	Blue artillery killed	Red artillery	Red artillery killed
1	3006.	0.	36.	0.
2	2972.	15.	4789.	6.
3	2963.	9.	4791.	41.
4	2950.	6.	4768.	27.
5	3103.	14.	4727.	19.
6	3531.	33.	4786.	24.
7	3609.	10.	4773.	30.
8	3772.	2.	4858.	16.
9	3772.	10.	4845.	35.
10	3847.	5.	4885.	94.
11	3931.	4.	4750.	59.
12	4063.	4.	4779.	34.
13	4093.	3.	4661.	23.
14	4004.	2.	4638.	31.
15	4065.	0.	4415.	19.
16	4087.	0.	4321.	26.
17	4086.	1.	4314.	17.
18	4077.	0.	4283.	36.
19	4087.	0.	4246.	14.
20	4088.	1.	4242.	35.
21	4150.	3.	4110.	28.
22	4153.	0.	4016.	22.
23	4144.	0.	4014.	23.
24	4133.	1.	3981.	26.
25	4131.	0.	3971.	23.
26	4128.	0.	3944.	21.
27	4131.	0.	3925.	33.
28	4090.	0.	3916.	32.
29	4108.	0.	3895.	13.
30	4106.	0.	3854.	20.
31	4081.	2.	3867.	19.
32	4092.	0.	3856.	14.
33	4080.	9.	3824.	7.

large-scale force comparisons in a NATO-Warsaw Pact context and regional balance issues in the Mideast, respectively.

No distinction is made in these data between surviving resources and newly arrived resources. If the data base could identify these two components of the resources, the estimation model could attempt to distinguish between them.

One avenue for research would be to attempt to estimate the weighting parameters rather than to assume them (as done in constructing Tables 5 and 7.) Such an estimation would require a substantially more complex curve fitting, or regression, process. This would essentially require the addition of four estimation parameters to the current model. To estimate these additional parameters would probably require more days of usable data than the 10 days used in the present treatment.

ESTIMATION OF PARAMETERS

Procedure

The parameters are chosen to minimize the sum of squared residuals of the actual and estimated attrition. Specifically, allowing n to index the observations $n = 2, \ldots, 11$, it is desired to minimize

Table 4. Data on combat manpower (infantry, armor, and artillery).

Day	Blue manpower	Blue casualties	Red manpower	Red casualties
1	351005.	458.	0.	0.
2	349247.	1589.	360716.	2191.
3	347915.	2383.	356818.	2423.
4	358321.	2085.	353529.	2015.
5	366495.	2175.	350750.	1993.
6	387342.	1389.	356278.	1985.
7	403289.	1174.	354297.	2084.
8	410817.	1905.	361684.	2046.
9	412811.	1548.	359353.	2468.
10	426360.	1608.	362904.	2685.
11	432094.	1527.	359750.	2538.
12	451316.	2320.	362611.	2504.
13	451724.	1376.	361023.	2544.
14	451291.	1277.	356892.	2121.
15	461189.	1005.	349900.	1682.
16	465334.	1042.	346100.	1844.
17	467620.	1159.	343134.	1550.
18	467801.	1004.	340875.	1788.
19	474562.	832.	338278.	1724.
20	474192.	1831.	334356.	1752.
21	481704.	2259.	328069.	2054.
22	480952.	1639.	321195.	1709.
23	478593.	1228.	322830.	1946.
24	475732.	1868.	324376.	1865.
25	475685.	1276.	322337.	1676.
26	475155.	1379.	320612.	1434.
27	472749.	1643.	319143.	1696.
28	472535.	1281.	319259.	1536.
29	468127.	1083.	317406.	1167.
30	467646.	1681.	316217.	1579.
31	466072.	1597.	314858.	1504.
32	464643.	2098.	313074.	1425.
33	455218.	1483.	310347.	1213.

Note: Casualties are killed and wounded, they do not include captured/missing in action or disease and nonbattle injuries.

$$\mathrm{SS} = \sum_{n=2}^{6} (\dot{B}_n - a\,d\,R_n^p\,B_n^q)^2 + \sum_{n=2}^{6} (\dot{R}_n - b(1/d)\,B_n^p\,R_n^q)^2 + \sum_{n=7}^{11} (\dot{B}_n - a\,(1/d)\,R_n^p\,B_n^q)^2 + \sum_{n=7}^{11} (\dot{R}_n - b\,d\,B_n^p\,R_n^q)^2.$$

Basically, the procedure is combinatoric. It evaluates

$$\mathrm{SS}(a_i, b_j, d_k, p_l, q_m)$$

for all combinations of $i = 1, \ldots, 5, j = 1, \ldots, 5, k = 1, \ldots, 3, l = 1, \ldots, 5$, and $m = 1, \ldots, 5$.

Table 5. Data on combat forces.

Day	Blue forces	Blue losses	Red forces	Red losses
1	558820.	478.	1440.	0.
2	555482.	2594.	577446.	2656.
3	553625.	3833.	571923.	4303.
4	562661.	3615.	567134.	3415.
5	576795.	4200.	563255.	3263.
6	644252.	3424.	570018.	3275.
7	665764.	1804.	566877.	3799.
8	681412.	2350.	578629.	2866.
9	683076.	2698.	576223.	4518.
10	698910.	2858.	580074.	6985.
11	715159.	2177.	570005.	5638.
12	746996.	2890.	572431.	4299.
13	748809.	1891.	565438.	3879.
14	744651.	1732.	559707.	3611.
15	757369.	1335.	543135.	2657.
16	761959.	1442.	535210.	3339.
17	766030.	1409.	531414.	2400.
18	767856.	1284.	528070.	3448.
19	775027.	1312.	524078.	2454.
20	774222.	2391.	519911.	3242.
21	785154.	2819.	508234.	3474.
22	783612.	1909.	496810.	2659.
23	781243.	1488.	498355.	2931.
24	776607.	2148.	498631.	3145.
25	777075.	1456.	496137.	2606.
26	777365.	1519.	493132.	2429.
27	775589.	1743.	491073.	3101.
28	772775.	1426.	490784.	2896.
29	769082.	1128.	487771.	1702.
30	768136.	1801.	485412.	2724.
31	764742.	1997.	484348.	2394.
32	763743.	2318.	482584.	2065.
33	752793.	2243.	478492.	1548.

Note: Forces are tanks, APCs, artillery, and combat manpower weighted by 20, 5, 40, and 1, respectively.

Example

As an example consider the estimation of parameters for the case of Model 1. The range of possibilities allowed is

$$a_1 = b_1 = 0.000\,000\,004,$$

$$a_2 = b_2 = 0.000\,000\,006,$$

$$a_3 = b_3 = 0.000\,000\,008,$$

$$a_4 = b_4 = 0.000\,000\,010,$$

$$a_5 = b_5 = 0.000\,000\,012,$$

Table 6. Data on total manpower.

Day	Blue manpower	Blue casualties	Red manpower	Red casualties
1	632105.	1468.	0.	0.
2	630557.	3062.	575838.	5590.
3	628985.	5712.	571301.	5559.
4	640969.	5093.	568508.	4711.
5	807140.	12101.	565173.	4332.
6	834136.	5334.	572181.	4351.
7	859906.	3197.	570711.	4582.
8	874600.	4815.	581177.	4531.
9	877247.	3730.	579660.	5351.
10	895976.	3857.	584610.	5609.
11	007490.	3635.	580731.	5563.
12	933045.	5411.	584551.	5526.
13	948024.	3596.	583610.	5751.
14	928230.	3435.	578737.	4511.
15	941188.	2934.	568768.	3900.
16	946424.	2743.	564548.	4076.
17	948226.	3022.	560993.	3635.
18	948379.	2773.	558214.	3898.
19	956144.	2631.	555741.	3821.
20	955821.	3580.	550854.	3892.
21	965135.	4899.	544031.	4283.
22	964928.	4093.	534885.	3767.
23	962193.	3388.	536481.	4169.
24	959776.	4627.	540896.	4076.
25	959011.	3928.	538328.	3756.
26	958799.	3725.	536719.	3466.
27	956330.	4002.	534764.	3732.
28	956090.	3502.	533256.	3967.
29	952030.	3590.	530919.	3199.
30	952210.	4189.	528237.	4026.
31	950879.	4277.	526387.	3866.
32	949508.	4477.	524150.	3744.
33	937500.	3600.	521038.	3219.

Note: Casualties are killed, wounded, captured/missing in action, and disease and nonbattle injuries.

$$d_1 = \tfrac{10}{10},$$

$$d_2 = \tfrac{10}{8},$$

$$d_3 = \tfrac{10}{6},$$

$$p_1 = q_1 = 0.8,$$

$$p_2 = q_2 = 0.9,$$

$$p_3 = q_3 = 1.0,$$

Table 7. Data on total forces.

Day	Blue forces	Blue losses	Red forces	Red losses
1	839920.	1488.	1440.	0.
2	836792.	4067.	792568.	6055.
3	834695.	7162.	786406.	7439.
4	845309.	6623.	782113.	6111.
5	1017440.	14126.	777678.	5602.
6	1091046.	7369.	785921.	5641.
7	1122381.	3827.	783291.	6297.
8	1145195.	5260.	798122.	5351.
9	1147512.	4880.	796530.	7401.
10	1168526.	5107.	801780.	9909.
11	1190555.	4285.	790986.	8663.
12	1228725.	5981.	794371.	7321.
13	1245109.	4111.	788025.	7086.
14	1221590.	3890.	781552.	6001.
15	1237368.	3264.	762003.	4875.
16	1243049.	3143.	753658.	5571.
17	1246636.	3272.	749273.	4485.
18	1248434.	3053.	745409.	5558.
19	1256609.	3111.	741541.	4551.
20	1255851.	4140.	736409.	5382.
21	1268585.	5459.	724196.	5703.
22	1267588.	4363.	710500.	4717.
23	1264843.	3648.	712006.	5154.
24	1260651.	4907.	715151.	5356.
25	1260401.	4108.	712128.	4686.
26	1261009.	3865.	709239.	4461.
27	1259170.	4102.	706694.	5137.
28	1256330.	3647.	704781.	5327.
29	1252985.	3635.	701284.	3734.
30	1252700.	4309.	697432.	5171.
31	1249549.	4677.	695877.	4756.
32	1248608.	4697.	693660.	4384.
33	1235075.	4360.	689183.	3554.

Note: Forces are tanks, APCs, artillery, and total manpower weighted by 20, 5, 40, and 1, respectively.

$$p_4 = q_4 = 1.1,$$

$$p_5 = q_5 = 1.2.$$

There are a total of $5 \times 5 \times 3 \times 5 \times 5 = 1875$ combinations of parameters.
The algorithm identifies the minimum sum of squared residuals

$$\mathrm{SS}(a_3, b_4, d_2, p_3, q_3).$$

Table 8 gives the sums of squared residuals for the selection of a_i and b_j corresponding to the minimum sum of squared residuals, namely,

$$a_3 = 0.000\,000\,008,$$

$$b_4 = 0.000\,000\,010,$$

Table 8. Sums of squared residuals for example.

	$a_3 = 0.000000008$, $b_4 = .000000010$					
	$q_1 = 0.8$	$q_2 = 0.9$	$q_3 = 1.0$	$q_4 = 1.1$	$q_5 = 1.2$	
$p_1 = 0.8$	0.273E+09	0.267E+09	0.244E+09	0.167E+09	0.259E+08	$d_1 = 10/10$
$p_2 = 0.9$	0.266E+09	0.243E+09	0.167E+09	0.247E+08	0.161E+10	
$p_3 = 1.0$	0.243E+09	0.166E+09	0.236E+08	0.162E+10	0.382E+11	
$p_4 = 1.1$	0.166E+09	0.226E+08	0.163E+10	0.384E+11	0.616E+12	
$p_5 = 1.2$	0.217E+08	0.164E+10	0.386E+11	0.620E+12	0.913E+13	
	0.273E+09	0.266E+09	0.241E+09	0.160E+09	0.166E+08	$d_2 = 10/8$
	0.266E+09	0.241E+09	0.160E+09	0.164E+08	0.182E+10	
	0.241E+09	0.159E+09	0.163E+08	0.186E+10	0.427E+11	
	0.158E+09	0.164E+08	0.189E+10	0.433E+11	0.690E+12	
	0.165E+08	0.193E+10	0.440E+11	0.699E+12	0.103E+14	
	0.272E+09	0.264E+09	0.236E+09	0.147E+09	0.405E+08	$d_3 = 10/6$
	0.264E+09	0.236E+09	0.146E+09	0.423E+08	0.301E+10	
	0.236E+09	0.145E+09	0.443E+08	0.308E+10	0.632E+11	
	0.144E+09	0.464E+08	0.317E+10	0.646E+11	0.100E+13	
	0.488E+08	0.325E+10	0.660E+11	0.102E+13	0.149E+14	

Mean daily attrition: 0.7027E+04 = 7027. Standard deviation = $\sqrt{0.1633E+08/10}$ = 1278.

displaying information for $d_1 = \frac{10}{10}$, $d_2 = \frac{10}{8}$, and $d_3 = \frac{10}{6}$, where for each case of d_k there is a 5 $\times$ 5 table for all combinations of p_l and q_m from 0.8 to 1.2.

The minimum is observed to be for $d_2 = \frac{10}{8}$, $p_3 = 1$, and $q_3 = 1$. The model with best fit of the parameters is thus

$$\dot{B} = .000\,000\,008\,(\tfrac{10}{8} \text{ or } \tfrac{8}{10})\, R^1 B^1,$$

$$\dot{R} = .000\,000\,010\,(\tfrac{8}{10} \text{ or } \tfrac{10}{8})\, B^1 R^1.$$

The mean of the average attrition suffered by both Blue and Red during days 2–11 is 0.7027E + 04, or 7027. The minimum sum of squared residuals is 0.1633E + 08, which yields a standard deviation of (0.1633E + 08/10)**0.5, or 1278.

Justification of Technique

The technique explained above can be characterized as brute force. A grid of parameters is specified. A search is performed over the grid. The minimum sum of squared residuals is identified. Then, if desired, a finer grid is specified which includes the parameters of the previous minimum, and the procedure is repeated. This does not guarantee that an optimal fit will be found. However, it does guarantee that the identified parameters are optimal over the options made available. In the example displayed in Table 8 the grid points have been chosen, after some experimentation, to illustrate the sensitivity of the sum of squared residuals as the parameters p, q, and d are varied.

Statistical techniques for regression rely on search procedures which can converge to local optima, perhaps far from a global optimum. Very few multiple nonlinear regression problems can be formulated as convex programming problems; see Bracken and McCor-

mick [3]. Thus, obtaining a globally optimal estimate of parameters can seldom be guaranteed.

The general significance of the parameter choices with respect to how well Lanchester square or Lanchester linear models fit the data is discussed in subsequent sections.

RESULTS

Results for the four models are as follows:

Model 1 —Combat Forces

$$\dot{B} = .000\,000\,008\,(\tfrac{10}{8} \text{ or } \tfrac{8}{10})\,R^1 B^1,$$

$$\dot{R} = .000\,000\,010\,(\tfrac{8}{10} \text{ or } \tfrac{10}{8})\,B^1 R^1.$$

Model 2—Total Forces

$$\dot{B} = .000\,000\,008\,(\tfrac{10}{8} \text{ or } \tfrac{8}{10})\,R^{0.8} B^{1.2},$$

$$\dot{R} = .000\,000\,008\,(\tfrac{8}{10} \text{ or } \tfrac{10}{8})\,B^{0.8} R^{1.2}.$$

Model 3—Combat Forces, No Tactical Parameter

$$\dot{B} = .000\,000\,008\,R^{1.3} B^{0.7},$$

$$\dot{R} = .000\,000\,010\,B^{1.3} R^{0.7}.$$

Model 4—Total Forces, No Tactical Parameter

$$\dot{B} = .000\,000\,008\,R^{1.2} B^{0.8},$$

$$\dot{R} = .000\,000\,008\,B^{1.2} R^{0.8}.$$

Tables 9–12 present sums of squared residuals followed by actual losses, estimated losses, and residuals for the best fits.

Each table is for the particular a_i, b_j associated with the best fit. Tables 9 and 10 include three values of d_k—$\frac{10}{10}$, $\frac{10}{8}$, and $\frac{10}{6}$. For each value of d_k is given a five-by-five array for five values each of p_l and q_m. Tables 11 and 12 are for no tactical parameter (or, equivalently, $d = 1$). Thus only one five-by-five array is given. Note that the grid spacing of p_l and q_m is not always the same. In Table 9 the grid variation is by 0.1, in Table 10 by 0.2, in Table 11 by 0.3, and in Table 12 by 0.2. The display given matches the computations performed. No attempt has been made to investigate the sensitivity of the choice of parameters to the choice of variation of grid size.

Tables 9–12 display average total losses and the standard deviation of the estimate. The standard deviations of the estimates are small compared to the mean. The tactical parameter helps to improve the quality of the fits, but even without the tactical parameter the fits are good as measured by the ratios of the standard deviations to the means.

Scanning the residuals across all four tables, it is observed that the residuals for Blue losses tend to be more negative and the residuals for Red losses tend to be more positive. Equivalently, estimated Blue losses are slightly higher than actual Blue losses, and estimated Red losses are slightly lower than actual Red losses. Total variation is used to choose the parameters of the best fit. An interpretation of the coefficients in this analysis is that the

Table 9. Model 1—Sums of squared residuals and NRL-804 actuals, estimates, and residuals for best fit.

Sums of squared residuals

	$a_3 = 0.000000008$ $b_4 = 0.000000010$					
	$q_1 = 0.8$	$q_2 = 0.9$	$q_3 = 1.0$	$q_4 = 1.1$	$q_5 = 1.2$	
$p_1 = 0.8$	0.273E+09	0.267E+09	0.244E+09	0.167E+09	0.259E+08	$d_1 = 10/10$
$p_2 = 0.9$	0.266E+09	0.243E+09	0.167E+09	0.247E+08	0.161E+10	
$p_3 = 1.0$	0.243E+09	0.166E+09	0.236E+08	0.162E+10	0.382E+11	
$p_4 = 1.1$	0.166E+09	0.226E+08	0.163E+10	0.384E+11	0.616E+12	
$p_5 = 1.2$	0.217E+08	0.164E+10	0.386E+11	0.620E+12	0.913E+13	
	0.273E+09	0.266E+09	0.241E+09	0.160E+09	0.166E+08	$d_2 = 10/8$
	0.266E+09	0.241E+09	0.160E+09	0.164E+08	0.182E+10	
	0.241E+09	0.159E+09	0.163E+08	0.186E+10	0.427E+11	
	0.158E+09	0.164E+08	0.189E+10	0.433E+11	0.690E+12	
	0.165E+08	0.193E+10	0.440E+11	0.699E+12	0.103E+14	
	0.272E+09	0.264E+09	0.236E+09	0.147E+09	0.405E+08	$d_3 = 10/6$
	0.264E+09	0.236E+09	0.146E+09	0.423E+08	0.301E+10	
	0.236E+09	0.145E+09	0.443E+08	0.308E+10	0.632E+11	
	0.144E+09	0.464E+08	0.317E+10	0.646E+11	0.100E+13	
	0.488E+08	0.325E+10	0.660E+11	0.102E+13	0.149E+14	

Actuals, estimates and residuals for best fit

Day	Blue losses	Estimated blue losses	Residual	Red losses	Estimated red losses	Residual
2	2594.	3208.	−614.	2656.	2566.	90.
3	3833.	3166.	667.	4303.	2533.	1770.
4	3615.	3191.	424.	3415.	2553.	862.
5	4200.	3249.	951.	3263.	2599.	664.
6	3424.	3672.	−248.	3275.	2938.	337.
7	1804.	2415.	−611.	3799.	4718.	−919.
8	2350.	2523.	−173.	2866.	4929.	−2063.
9	2698.	2519.	179.	4518.	4920.	−402.
10	2858.	2595.	263.	6985.	5068.	1917.
11	2177.	2609.	−432.	5638.	5096.	542.

Average total losses = 7027. Standard deviation = 1278.

method results in individual effectiveness of Red forces being slightly overestimated and individual effectiveness of Blue forces being slightly underestimated.

INTERPRETATION OF RESULTS

Lanchester Linear Result

The first main interpretation of results is that the Lanchester linear equation fits the campaign.

Model 1, for combat forces, with the tactical parameter $d = \frac{10}{8}$ (attrition to the defender is $\frac{10}{8}$ of that to the attacker), is exactly Lanchester linear. Referring to the top of Table 9, note that the fit along the diagonal is not tight, so the pair $p = 0.8$, $q = 1.2$ or the pair $p =$

Table 10. Model 2—Sums of squared residuals and actuals, estimates, and residuals for best fit.

Sums of squared residuals

$a_3 = 0.000000008$ $b_3 = 0.00000008$

	$q_1 = 0.6$	$q_2 = 0.8$	$q_3 = 1.0$	$q_4 = 1.2$	$q_5 = 1.4$	
$p_1 = 0.6$	0.965E+09	0.965E+09	0.958E+09	0.858E+09	0.159E+09	$d_1 = 10/10$
$p_2 = 0.8$	0.965E+09	0.958E+09	0.858E+09	0.137E+09	0.192E+12	
$p_3 = 1.0$	0.958E+09	0.858E+09	0.124E+09	0.189E+12	0.520E+14	
$p_4 = 1.2$	0.857E+09	0.119E+09	0.188E+12	0.516E+14	0.127E+17	
$p_5 = 1.4$	0.123E+09	0.192E+12	0.520E+14	0.127E+17	0.308E+19	
	0.965E+09	0.964E+09	0.957E+09	0.852E+09	0.990E+08	$d_2 = 10/8$
	0.964E+09	0.957E+09	0.851E+09	0.938E+08	0.203E+12	
	0.957E+09	0.850E+09	0.974E+08	0.205E+12	0.557E+14	
	0.849E+09	0.110E+09	0.211E+12	0.568E+14	0.137E+17	
	0.133E+09	0.222E+12	0.589E+14	0.141E+17	0.339E+19	
	0.965E+09	0.964E+09	0.956E+09	0.836E+09	0.176E+09	$d_3 = 10/6$
	0.964E+09	0.956E+09	0.835E+09	0.193E+09	0.285E+12	
	0.956E+09	0.833E+09	0.223E+09	0.296E+12	0.775E+14	
	0.831E+09	0.268E+09	0.312E+12	0.810E+14	0.193E+17	
	0.329E+09	0.334E+12	0.859E+14	0.204E+17	0.483E+19	

Actuals, estimates and residuals for best fit

Day	Blue losses	Estimated blue losses	Residual	Red losses	Estimated red losses	Residual
2	4067.	6705.	−2638.	6055.	4199.	1856.
3	7162.	6643.	519.	7439.	4151.	3288.
4	6623.	6715.	−92.	6111.	4166.	1945.
5	14126.	8349.	5777.	5602.	4799.	803.
6	7369.	9156.	−1787.	5641.	5139.	502.
7	3827.	6046.	−2219.	6297.	8181.	−1884.
8	5260.	6288.	−1028.	5351.	8503.	−3152.
9	4880.	6293.	−1413.	7401.	8497.	−1096.
10	5107.	6465.	−1358.	9909.	8689.	1220.
11	4285.	6541.	−2256.	8663.	8678.	−15.

Average total losses = 13,120. Standard deviation = 3063.

1.2, $q = 0.8$ would be about the same. However, these fits would still be close to Lanchester linear.

Model 2, for total forces, with the tactical parameter $d = \frac{10}{8}$, yields the best fit for $p = 0.8$ and $q = 1.2$, but the fit is almost the same for $p = 1$ and $q = 1$.

Model 3, for combat forces, without the tactical parameter, has its best fit with $p = 1.3$ and $q = 0.7$. Table 11 also gives an indication of sensitivity, however, because the grid is more widely defined. Though the fits are about the same from $p = 1.3$, $q = 0.7$ to $p = 0.7$, $q = 1.3$, they are much worse outside of that range on the diagonal (namely, for $p = 1.7$, $q = 0.4$ and $p = 0.4$, $q = 1.7$). This tends to support the conclusion that the Lanchester linear model describes the campaign. Significant variations along the diagonal towards $p = 1$, $q = 0$ (Lanchester square model) or toward $p = 0$, $q = 1$ (logarithmic model) do not fit the Ardennes campaign.

Model 4, for total forces, without the tactical parameter, has its best fit with $p = 1.2$ and

Table 11. Model 3—Sums of squared residuals and actuals, estimates, and residuals for best fit.

Sums of squared residuals

$a_3 = 0.000000008$ $b_4 = 0.000000010$

	$q_1 = 0.4$	$q_2 = 0.7$	$q_3 = 1.0$	$q_4 = 1.3$	$q_5 = 1.7$
$p_1 = 0.4$	0.275E+09	0.275E+09	0.275E+09	0.267E+09	0.162E+10
$p_2 = 0.7$	0.275E+09	0.275E+09	0.267E+09	0.272E+08	0.899E+13
$p_3 = 1.0$	0.275E+09	0.266E+09	0.236E+08	0.614E+12	0.269E+17
$p_4 = 1.3$	0.266E+09	0.208E+08	0.623E+12	0.190E+16	0.799E+20
$p_5 = 1.7$	0.172E+10	0.947E+13	0.279E+17	0.816E+20	0.343E+25

Actuals, estimates, and residuals for best fit

Day	Blue losses	Estimated blue losses	Residual	Red losses	Estimated red losses	Residual
2	2594.	2596.	−2.	2656.	3171.	−515.
3	3833.	2558.	1275.	4303.	3136.	1167.
4	3615.	2559.	1056.	3415.	3183.	232.
5	4200.	2581.	1619.	3263.	3272.	−9.
6	3424.	2832.	592.	3275.	3810.	−535.
7	1804.	2877.	−1073.	3799.	3961.	−162.
8	2350.	3003.	−653.	2866.	4141.	−1275.
9	2698.	2992.	−294.	4518.	4142.	376.
10	2858.	3067.	−209.	6985.	4287.	2698.
11	2177.	3047.	−870.	5638.	4364.	1274.

Average total losses = 7027. Standard deviation = 1444.

$q = 0.8$. As in the other models, the fit does not seem to be too tight, but it should be expected that parameter variations far along the diagonal would lead to the same behavior as in Model 3.

From the point of view of defense planning and programming, the quantity-quality trade-offs suggested by the present analysis of the Ardennes campaign tend to be equal. Bracken [2] discusses this issue in the context of the defense budget (which was one of the motivations of the present research on the Ardennes campaign.) Modern warfare, leading to concentration of forces and to the Lanchester square law as discussed in Lanchester [23], was apparently not present in the Ardennes campaign in 1944.

Combat Forces and Total Forces

The second main interpretation of results is that the individual effectiveness parameters depend upon whether combat forces in the campaign or total forces in the campaign are included. When combat forces are included the Allies have greater individual effectiveness than the Germans, while when total forces are included the individual effectiveness is identical.

This is an intuitively appealing result. The two sides were organized quite differently, with the Allied combat forces forming a much smaller proportion of the Allied total forces than the German combat forces as a proportion of the German total forces. The Allied support forces caused the individual effectiveness of the Allied combat forces to be greater than the individual effectiveness of the German combat forces. However, when total forces

Table 12. Model 4—Sums of squares residuals and actuals, estimates, and residuals for best fit.

Sums of squared residuals

	$a_3 = 0.000000008$ $b_3 = 0.000000008$				
	$q_1 = 0.6$	$q_2 = 0.8$	$q_3 = 1.0$	$q_4 = 1.2$	$q_5 = 1.4$
$p_1 = 0.6$	0.965E+09	0.965E+09	0.958E+09	0.858E+09	0.159E+09
$p_2 = 0.8$	0.965E+09	0.958E+09	0.858E+09	0.137E+09	0.192E+12
$p_3 = 1.0$	0.958E+09	0.858E+09	0.124E+09	0.189E+12	0.520E+14
$p_4 = 1.2$	0.857E+09	0.119E+09	0.188E+12	0.516E+14	0.127E+17
$p_5 = 1.4$	0.123E+09	0.192E+12	0.520E+14	0.127E+17	0.308E+19

Actuals, estimates and residuals for best fit

Day	Blue losses	Estimated blue losses	Residual	Red losses	Estimated red losses	Residual
2	4067.	5248.	−1181.	6055.	5364.	691.
3	7162.	5189.	1973.	7439.	5314.	2125.
4	6623.	5207.	1416.	6111.	5372.	739.
5	14126.	5999.	8127.	5602.	6679.	−1077.
6	7369.	6424.	945.	5641.	7325.	−1684.
7	3827.	6545.	−2718.	6297.	7558.	−1261.
8	5260.	6803.	−1543.	5351.	7860.	−2509.
9	4880.	6797.	−1917.	7401.	7866.	−465.
10	5107.	6951.	−1844.	9909.	8082.	1827.
11	4285.	6942.	−2657.	8663.	8176.	487.

Average total losses = 13,120. Standard deviation = 3455.

are included, the individual effectiveness of the Allied and German forces are identical. This confirms intuition, for the Allies and Germans brought similar individual human and physical capital endowments to the warfare situation. The military organizations were configured quite differently, but resulted in identical individual effectiveness.

LIMITATIONS AND DIRECTIONS FOR FURTHER RESEARCH

The present models are homogeneous. They involve assigning reasonable, but subjective, weights to tanks, armored personnel carriers, artillery, and manpower, and then estimating parameters for the resulting homogeneous forces. Heterogeneous models would not have this limitation, but they would have many more parameters. Graham [13] formulates such models for American Civil War battles and estimates their parameters. Perhaps using a variation of his approach (with necessarily more parameters, because there are more types of forces in the twentieth century) would yield reasonably good heterogeneous models for the Ardennes campaign.

Engaged forces are implied by the derivation of the Lanchester models. The differential, or difference, equations of the Lanchester models (see, for instance, Taylor [29]) include the physical interactions of shooters and targets rather than the combination of engaged forces and other than engaged forces, which the present article defines as combat forces or total forces. This is a limitation of the present article in that the theory and empirical work do not strictly correspond. However, the level of aggregation of the present article is still of interest, for many decisions are made on the basis of forces in being, with the knowledge

that only a fraction of these forces will actually engage. This area of model definition and scope might be useful for further investigation.

The parameter estimation procedure of the present article is quite straightforward. It examines a grid of 1875 five-dimensional vectors of parameters, finds where in the space the sum of squared differences of actual and estimated attrition is minimized, and displays 5×5 arrays of sums of squared deviations for variations in the exponent parameters, and for three tactical parameters, given the best pair of effectiveness parameters. Variations in all five parameter dimensions are of interest and have not been thoroughly explored. No local minimum problems have been observed, but they might be present and this consideration should be investigated. Finally, detailed variations in parameters to obtain best fits, or at least tighter fits, should be explored.

The models presented in this article do not include the effects of air forces. Data for air forces are contained in the data base and should be examined in the context of an expanded model.

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